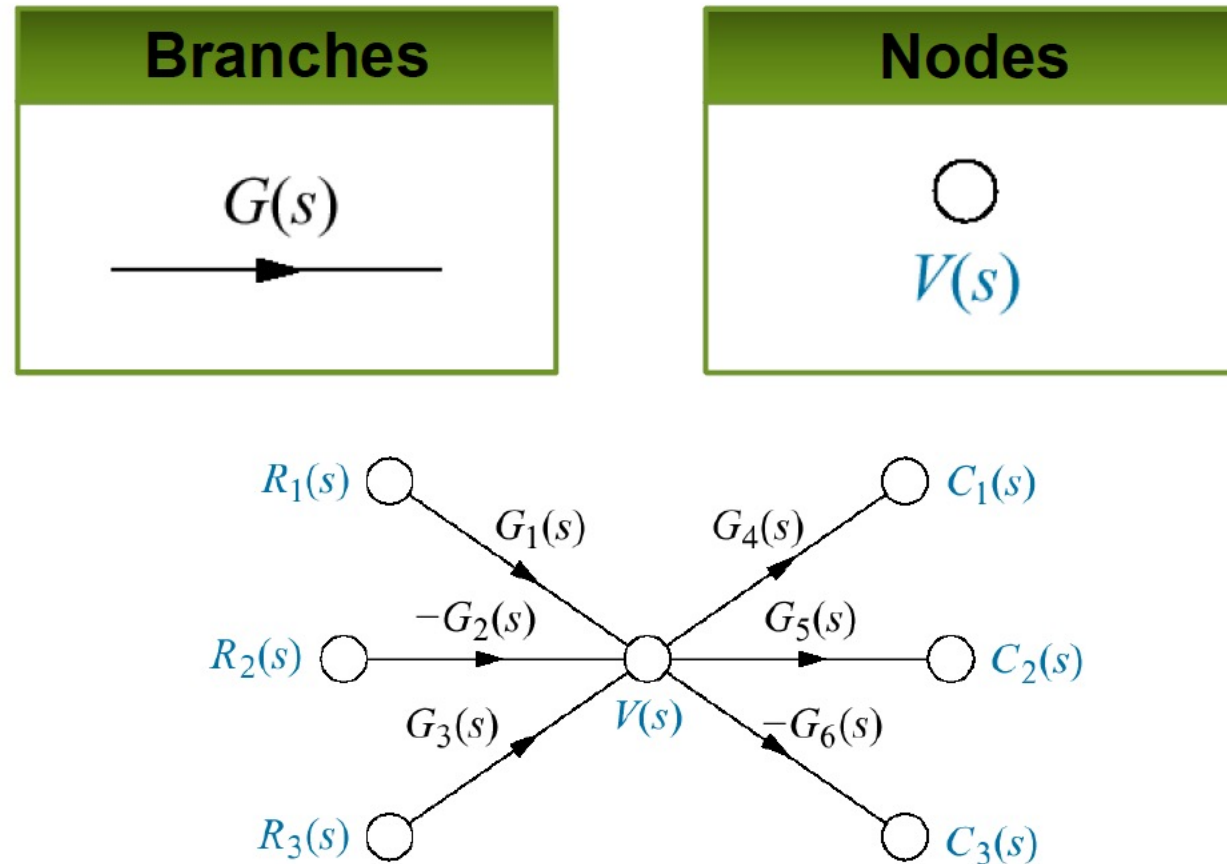


Signal Flow Graphs

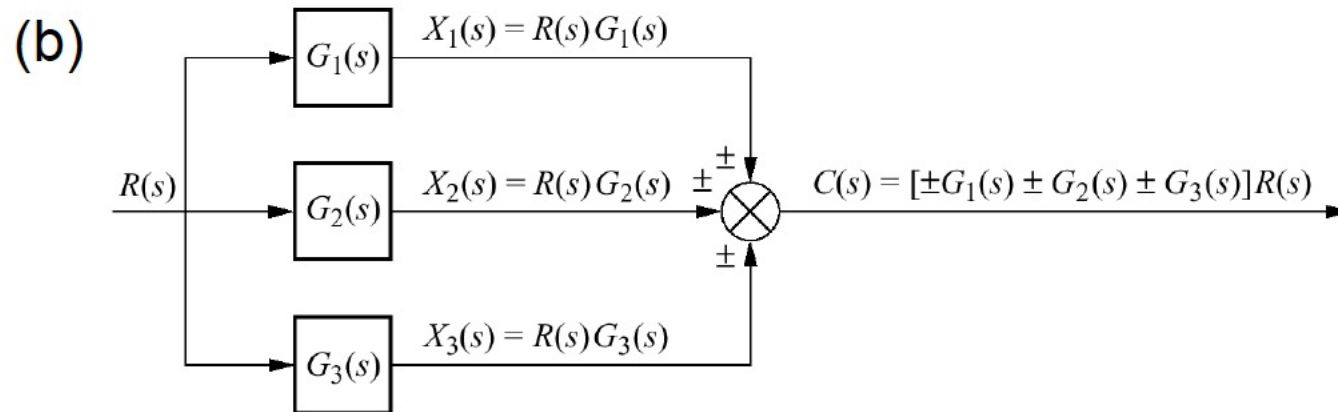
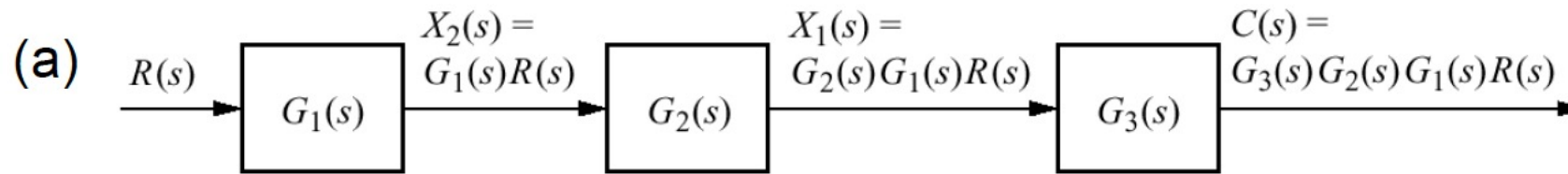
- Alternative to block diagram;
- Consists only **branches** (systems), and **nodes** (signals)



Signal Flow Graphs

Example 3-1

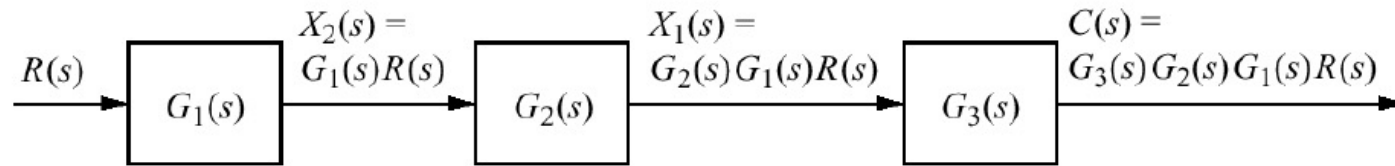
Convert the cascaded, parallel, and feedback forms of the block diagrams shown in figures below, respectively, into signal-flow graphs.



Signal Flow Graphs

Solution 3-1

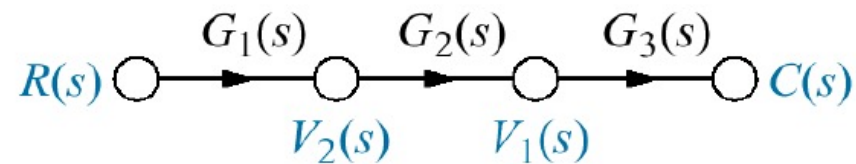
(a)



- draw the signal nodes for the system:



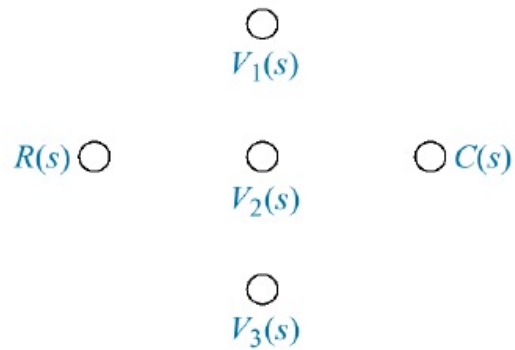
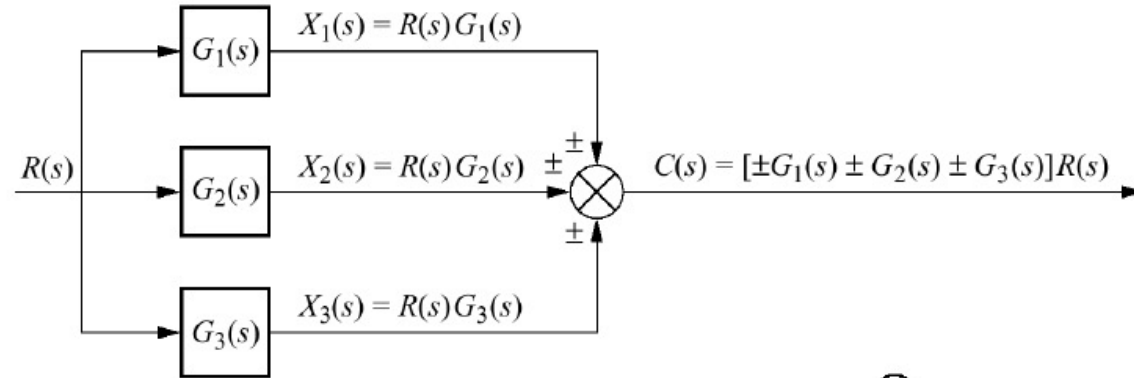
- interconnect the signal codes with system branches :



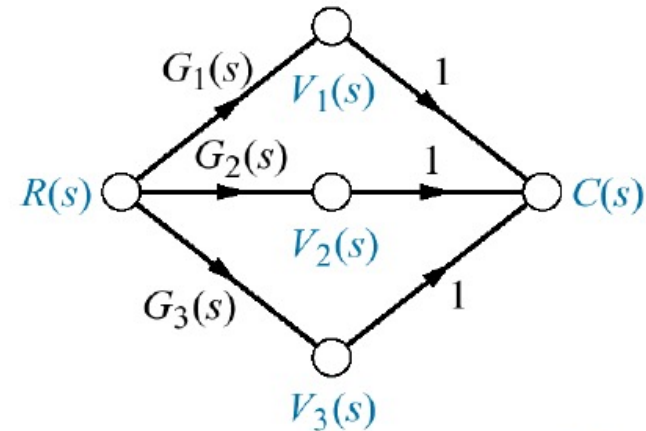
Signal Flow Graphs

Solution 3-1

(b)



Draw the signal nodes

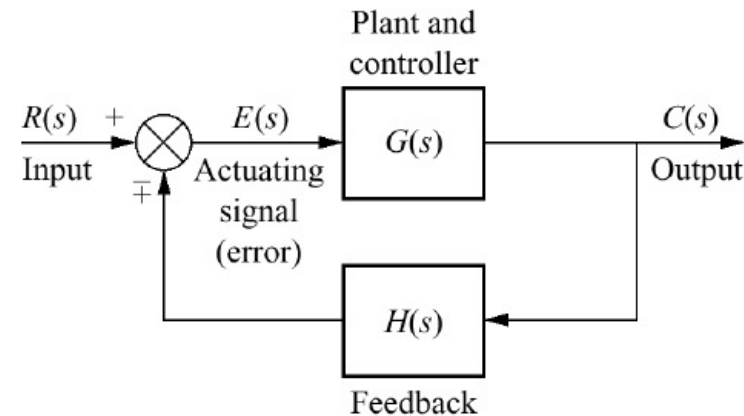


Interconnect the signal nodes with system branches

Signal Flow Graphs

Example 3-2

Convert the cascaded, parallel, and feedback forms of the block diagrams shown in figure below into signal-flow graphs.

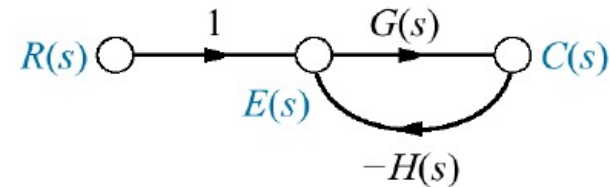


Solution 3-2

$R(s)$ ○

○
 $E(s)$

○ $C(s)$



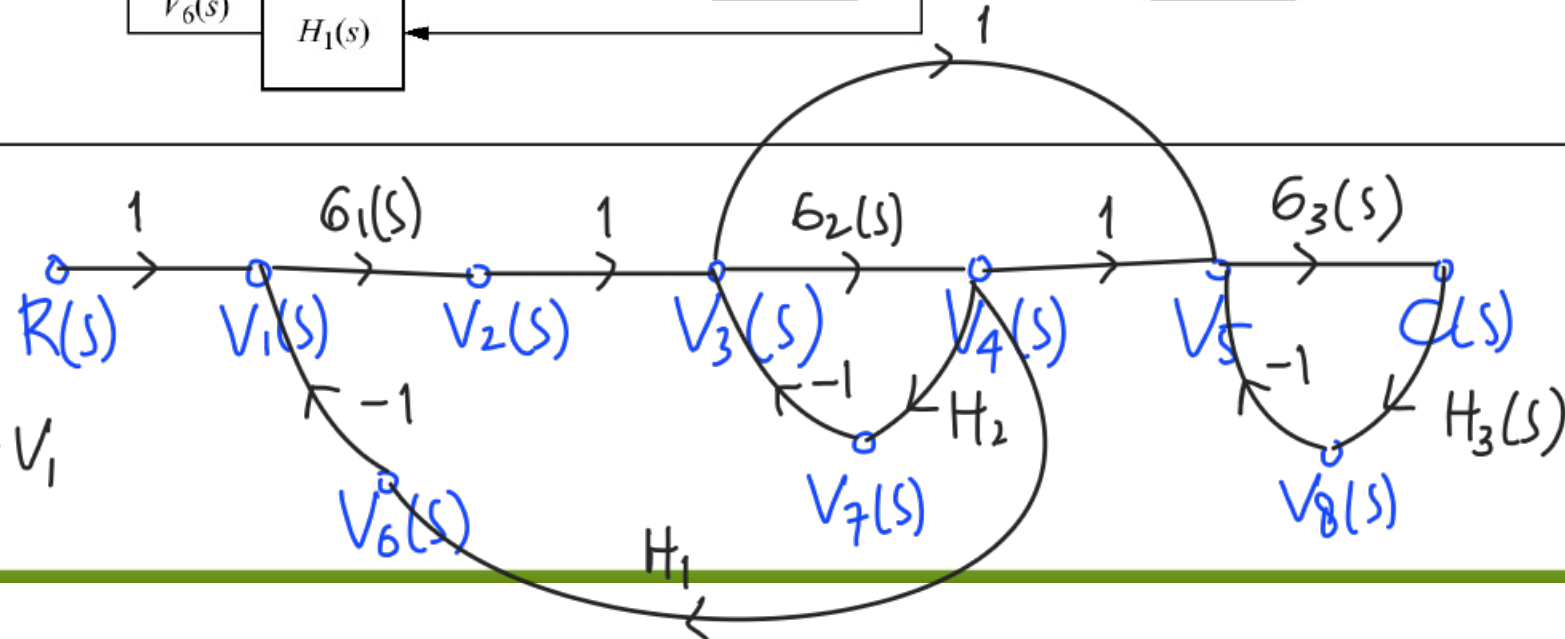
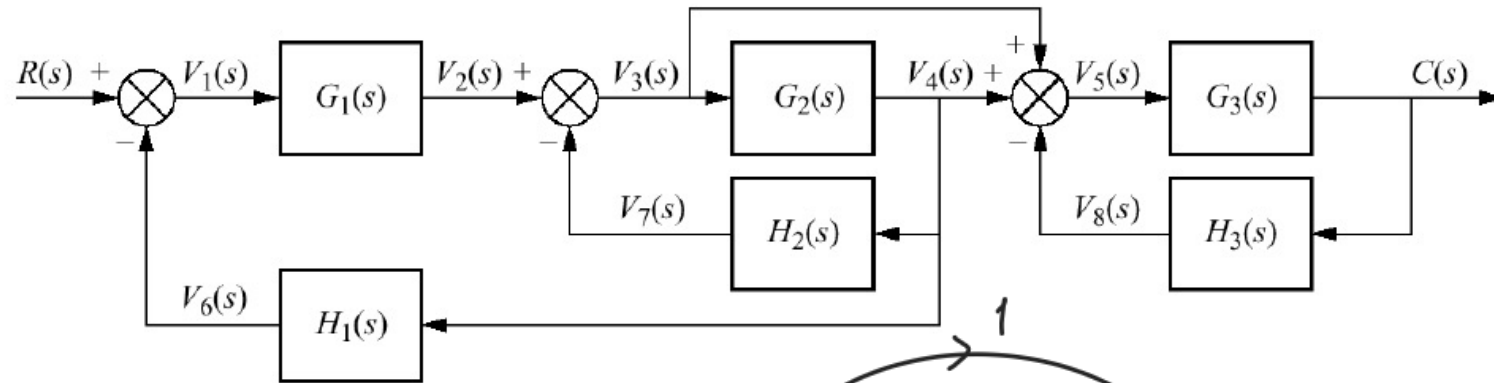
Signal Flow Graphs

Forward path : 2 path

- $R \rightarrow V_1 \rightarrow V_2 \rightarrow V_3 \rightarrow V_4 \rightarrow V_5 \rightarrow C$
- $R \rightarrow V_1 \rightarrow V_2 \rightarrow V_3 \rightarrow V_5 \rightarrow C$

Exercise 3-1

Convert the block diagram of figure below to a signal-flow graph.



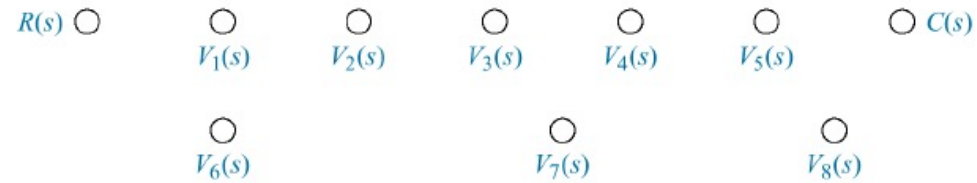
Loop : 3 loop

- $V_5 \rightarrow C \rightarrow V_5$
- $V_3 \rightarrow V_4 \rightarrow V_3$
- $V_1 \rightarrow V_2 \rightarrow V_3 \rightarrow V_4 \rightarrow V_1$

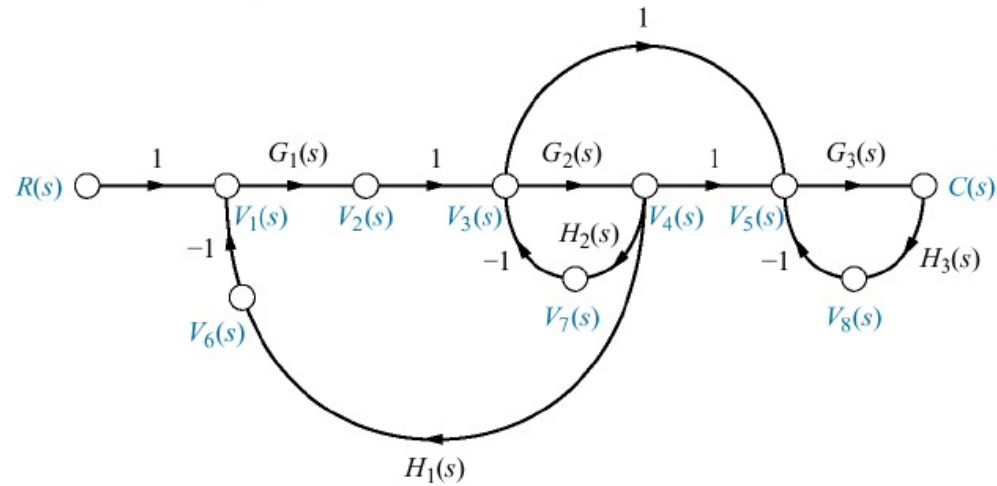
Signal Flow Graphs

Solution Ex 3-1

Draw the signal nodes



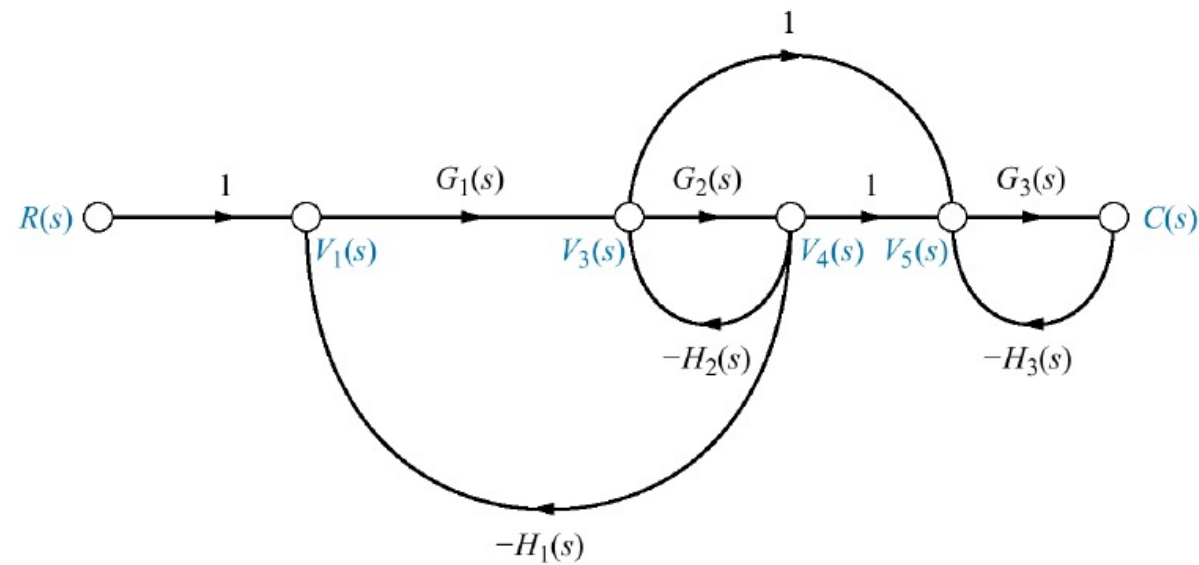
Interconnect the signal nodes with system branches



Signal Flow Graphs

Solution Ex 3-1

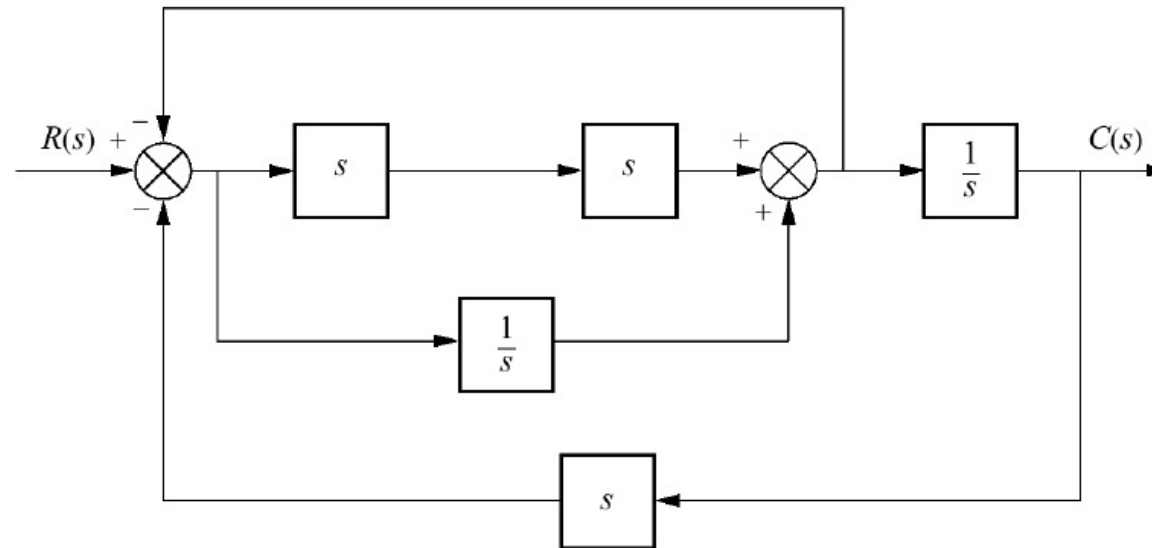
Simplify the signal-flow graph



Signal Flow Graphs

Exercise 3-2

Convert the block diagram of figure below to a signal-flow graph.



Mason's Rule

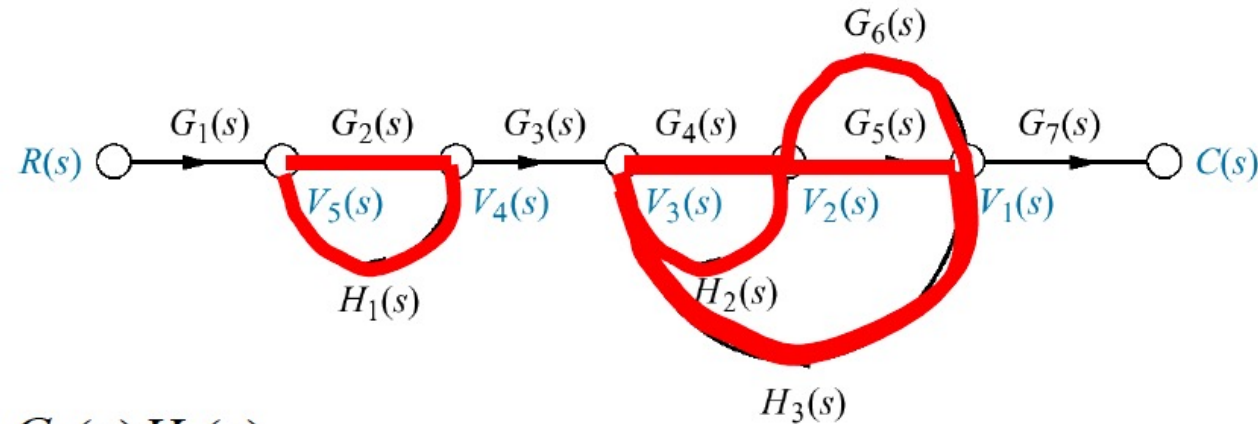
- A technique to reduce a signal-flow graph to a single transfer function
- Requires the application of one formula.
- The transfer function, $C(s)/R(s)$, of a system represented by a signal-flow graph is

$$G(s) = \frac{C(s)}{R(s)} = \frac{\sum_k T_k \Delta_k}{\Delta}$$

- ❑ k = number of forward path
- ❑ T_k = the k th forward path gain
- ❑ $\Delta = 1 - (\sum \text{loop gains}) + (\sum \text{non-touching loop gains taken two at a time}) - (\sum \text{non-touching loop gains taken three at a time})$.
- ❑ $\Delta_k = 1 - (\text{loop-gain which does not touch the forward path})$

Mason's Rule – definitions

- **Loop gain** – the product of branch gains found by traversing a path that starts at a node and ends at the same node



L1. $G_2(s)H_1(s)$

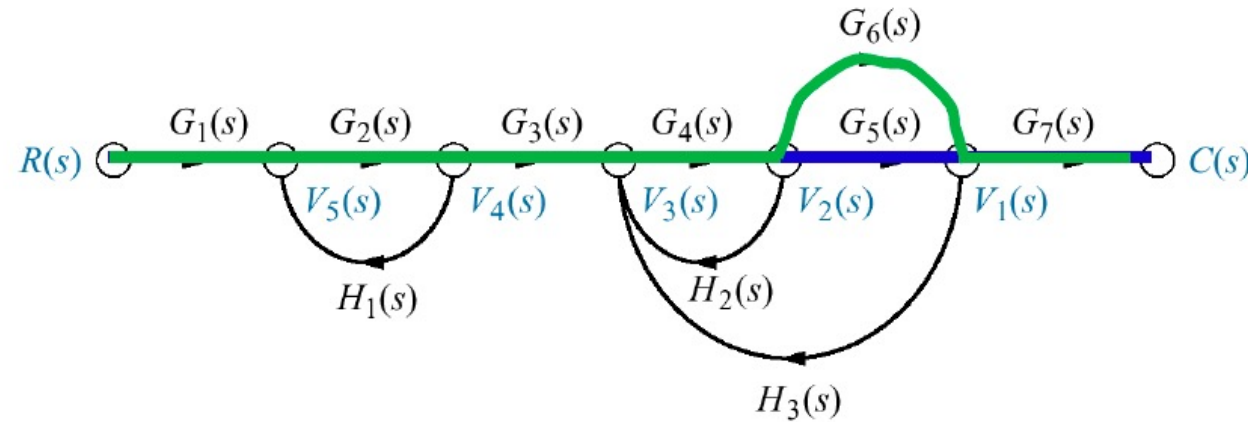
L2. $G_4(s)H_2(s)$

L3. $G_4(s)G_5(s)H_3(s)$

L4. $G_4(s)G_6(s)H_3(s)$

Mason's Rule – definitions

- **Forward path** – the product of gains found by traversing a path from the input node to the output node of the signal-flow graph

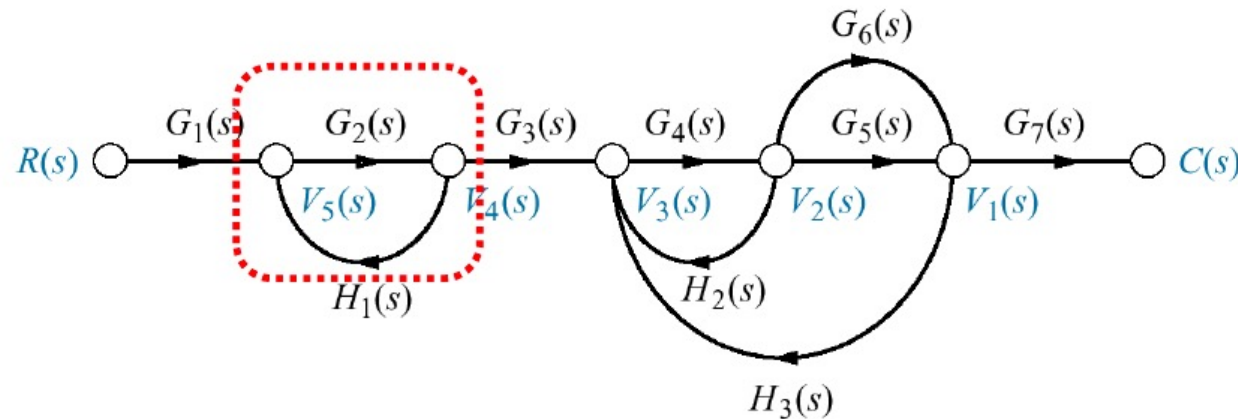


$$\text{F1. } G_1(s)G_2(s)G_3(s)G_4(s)G_5(s)G_7(s) \quad \Delta_1 = 1$$

$$\text{F2. } G_1(s)G_2(s)G_3(s)G_4(s)G_6(s)G_7(s) \quad \Delta_2 = 1$$

Mason's Rule – definitions

- **Non-touching loop** – Loops that do not have any nodes in common.
- **Non-touching loop gain** – The product of loop gains from non-touching loops taken two, three, four, or more at a time.



$$\text{NT1. } [G_2(s)H_1(s)][G_4(s)H_2(s)]$$

$$\text{NT2. } [G_2(s)H_1(s)][G_4(s)G_5(s)H_3(s)]$$

$$\text{NT3. } [G_2(s)H_1(s)][G_4(s)G_6(s)H_3(s)]$$

$$G(s) = \frac{T_1\Delta_1 + T_2\Delta_2}{\Delta},$$

$$T_1 = G_1 G_2 G_3 G_4 G_5 G_7, \Delta_1 = 1.$$

$$T_2 = G_1 G_2 G_3 G_4 G_6 G_7, \Delta_2 = 1.$$

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4) + (NT_1 + NT_2 + NT_3)$$

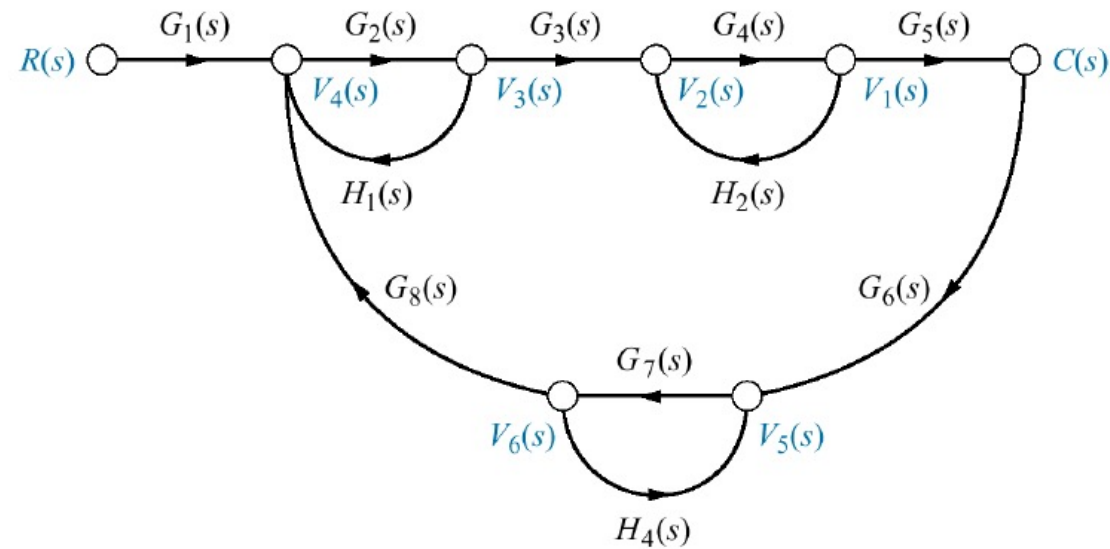
$$G(s) = \frac{F_1\Delta_1 + F_2\Delta_2}{\Delta}$$

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4) + (L_1L_2 + L_1L_3 + L_1L_4)$$

Mason's Rule

Example 3-3

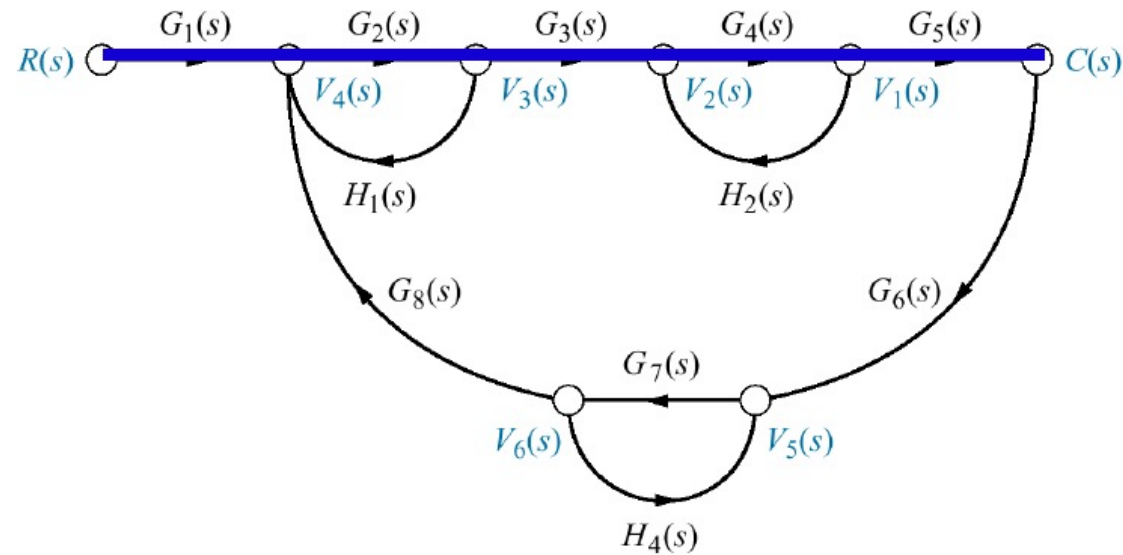
Find the transfer function, $C(s)/R(s)$, for the signal-flow graph in figure below



Mason's Rule

Solution 3-3

First, identify the forward-path gains.

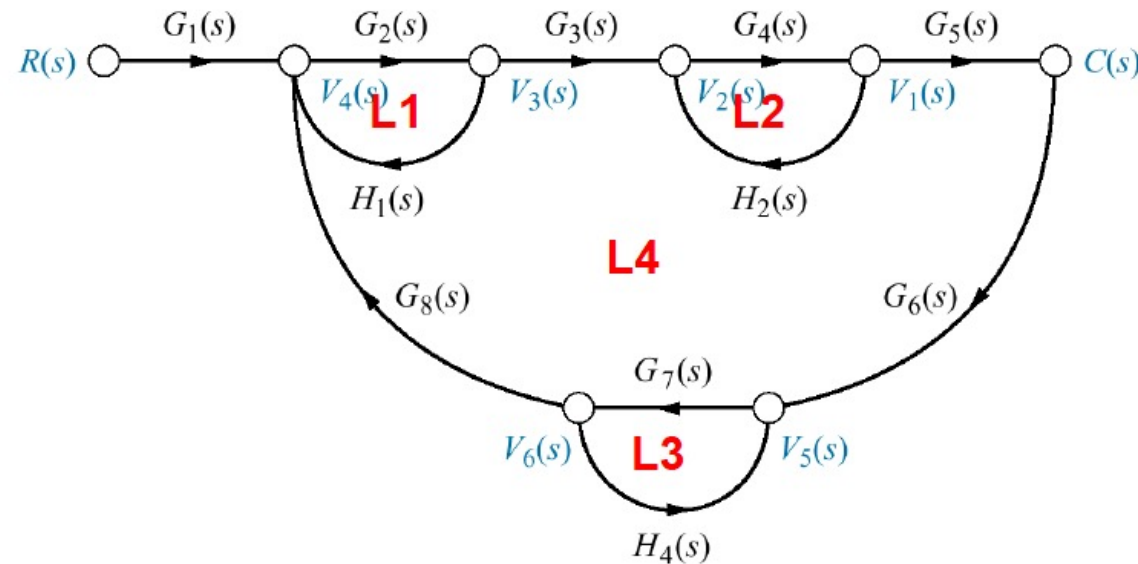


F1. $G_1(s)G_2(s)G_3(s)G_4(s)G_5(s)$

Mason's Rule

Solution 3-3

Second, identify the loop gains.



$$L1. G_2(s)H_1(s)$$

$$L3. G_7(s)H_4(s)$$

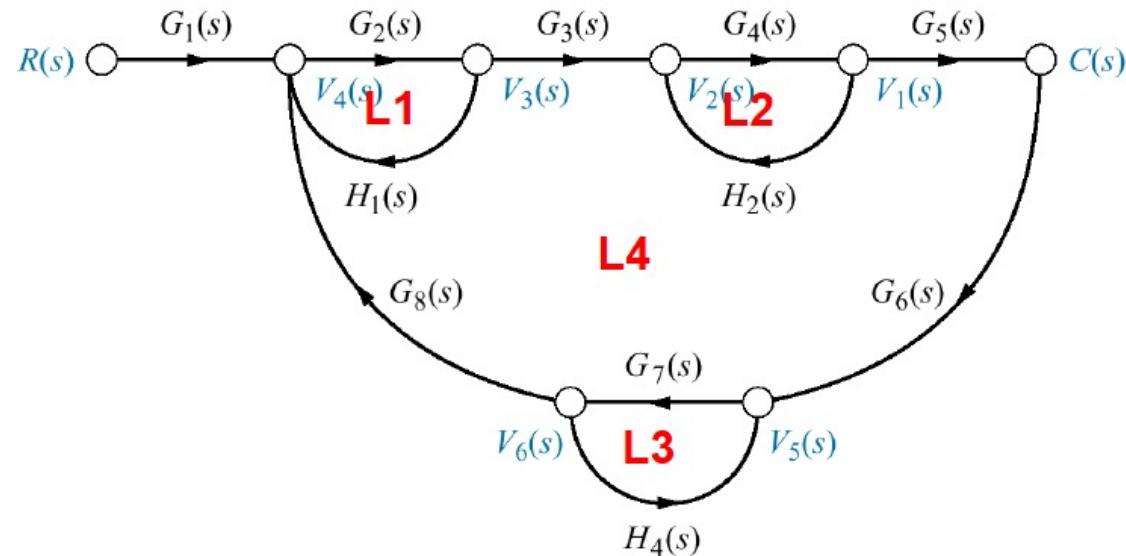
$$L2. G_4(s)H_2(s)$$

$$L4. G_2(s)G_3(s)G_4(s)G_5(s)G_6(s)G_7(s)G_8(s)$$

Mason's Rule

Solution 3-3

Third, identify the non-touching loops taken two at a time.



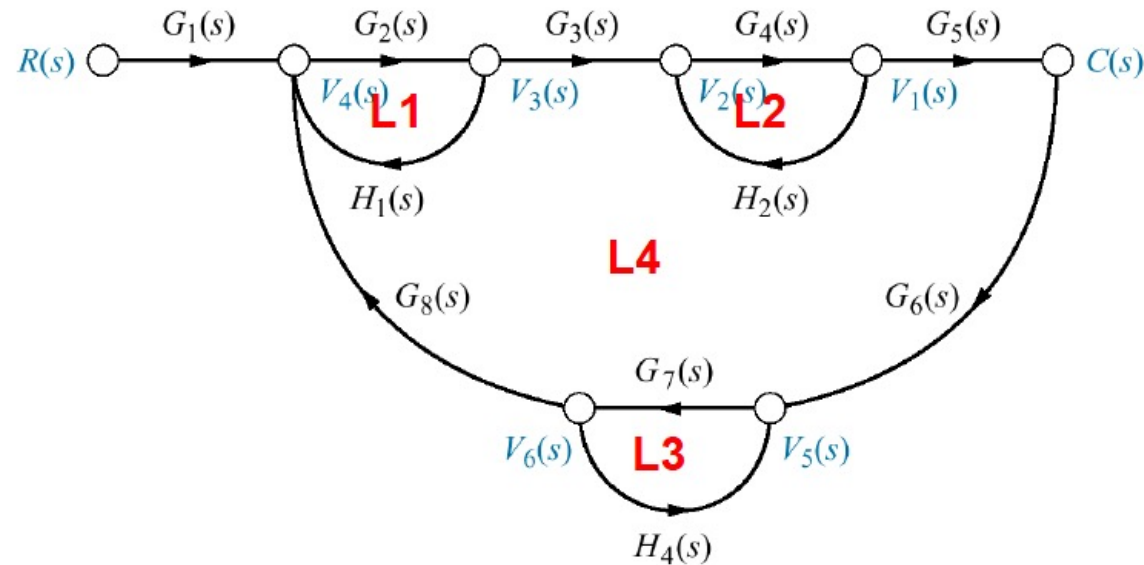
L1 and L2: $G_2(s)H_1(s)G_4(s)H_2(s)$ L2 and L3: $G_4(s)H_2(s)G_7(s)H_4(s)$

L1 and L3: $G_2(s)H_1(s)G_7(s)H_4(s)$

Mason's Rule

Solution 3-3

Finally, identify the non-touching loops taken three at a time.



L1, L2, L3: $G_2(s)H_1(s)G_4(s)H_2(s)G_7(s)H_4(s)$

Mason's Rule

Solution 3-3

Now, form the Δ and Δk .

$$\begin{aligned}\Delta = & 1 - [G_2(s)H_1(s) + G_4(s)H_2(s) \\ & + G_7(s)H_4(s) + G_2(s)G_3(s)G_4(s)G_5(s)G_6(s)G_7(s)G_8(s)] \\ & + [G_2(s)H_1(s)G_4(s)H_2(s) + G_2(s)H_1(s)G_7(s)H_4(s) \\ & + G_4(s)H_2(s)G_7(s)H_4(s)] \\ & - [G_2(s)H_1(s)G_4(s)H_2(s)G_7(s)H_4(s)]\end{aligned}$$

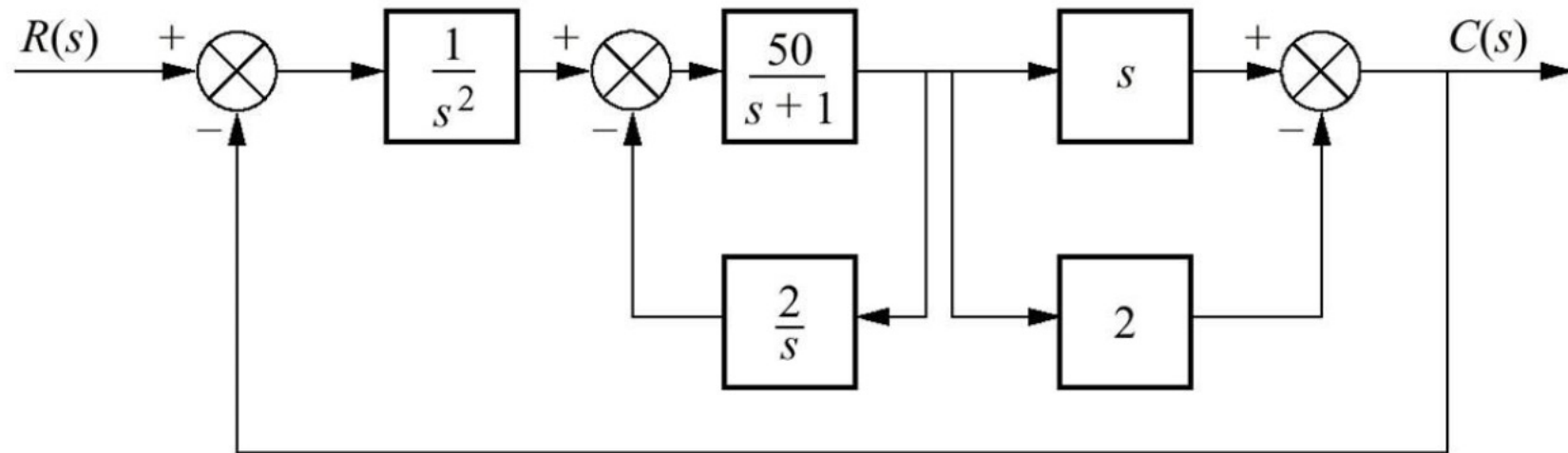
Δk is formed by [1 – loop does not touch forward path]

$$\Delta_1 = 1 - G_7(s)H_4(s)$$

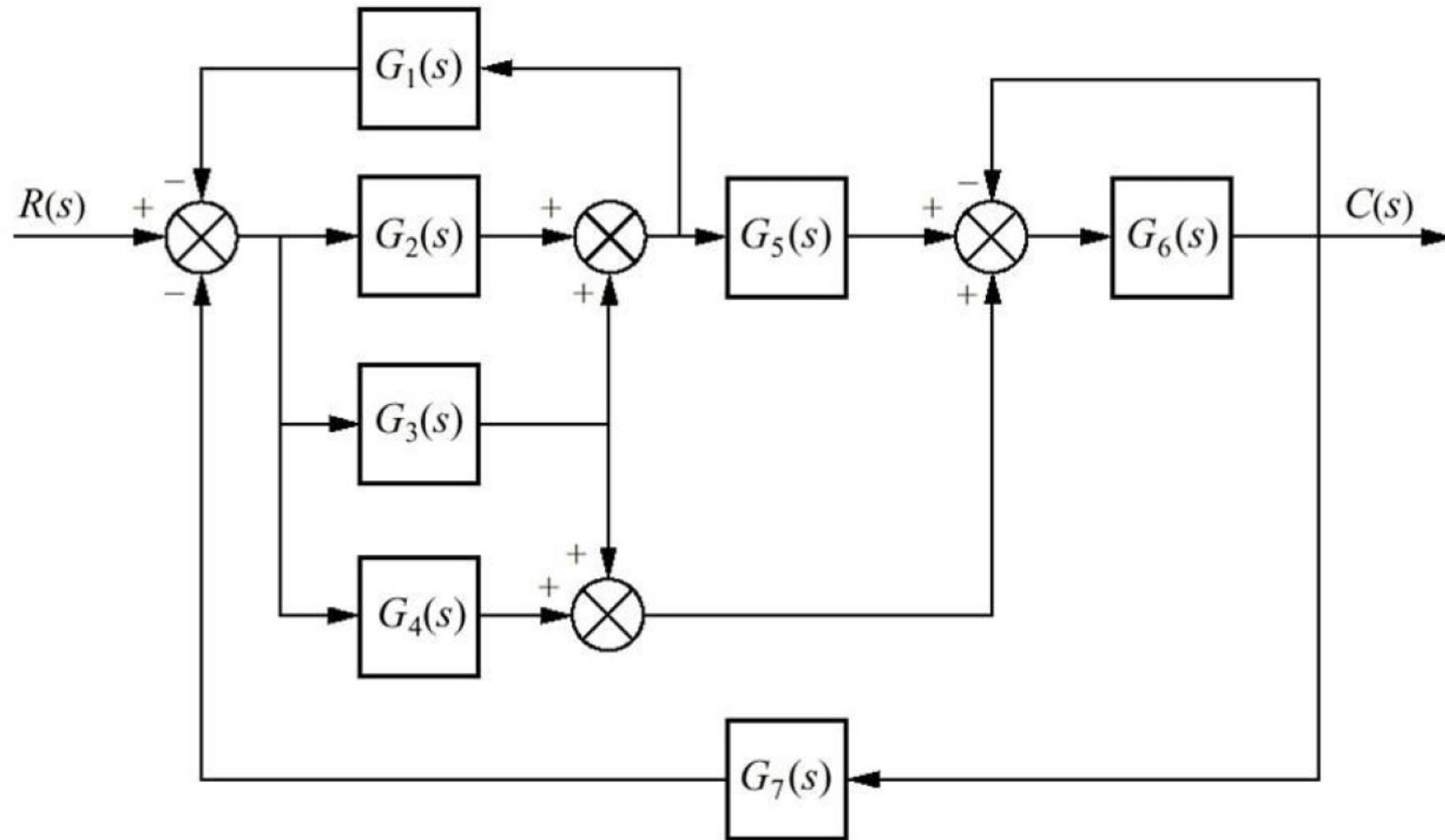
Latihan

1. Label signals and draw a signal-flow graph for each of the block diagrams shown in the following problems:

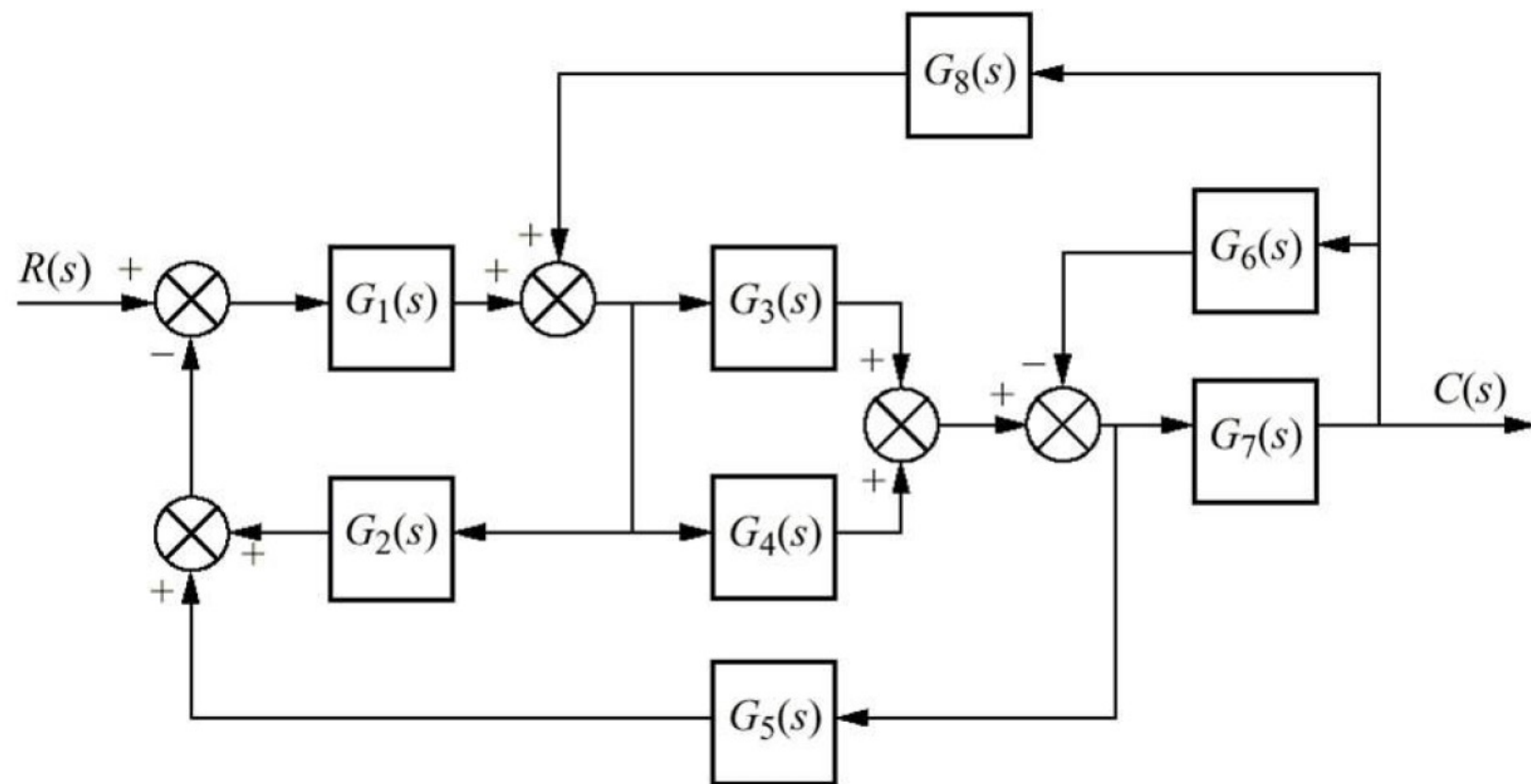
(a)



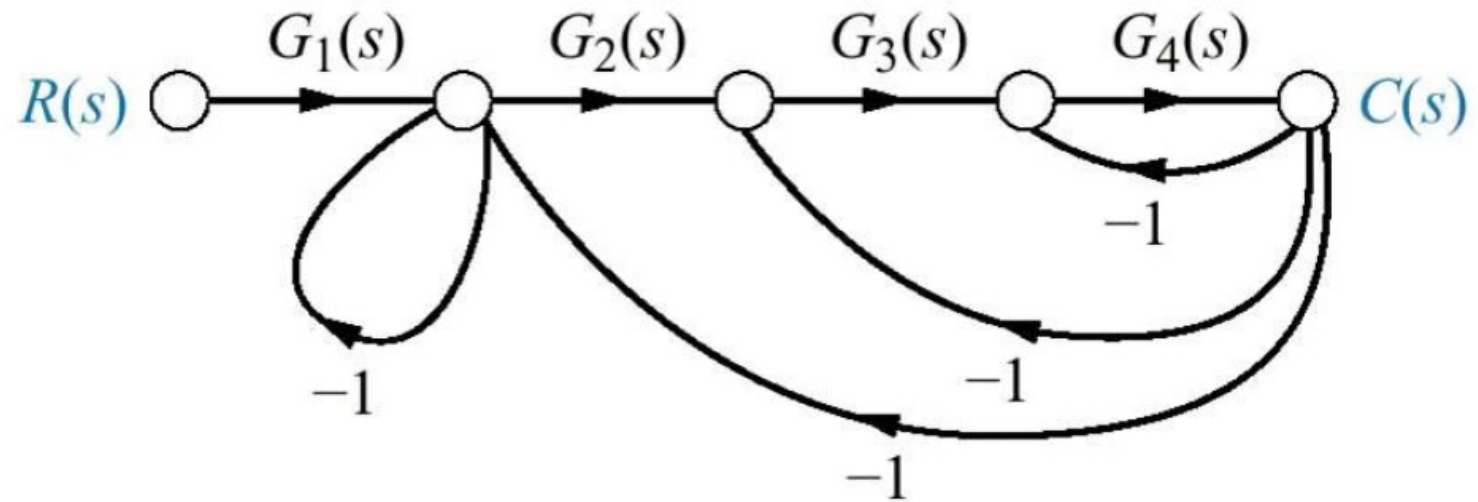
(b)



(c)



2. Using Mason's rule, find the transfer function, $T(s) = C(s) / R(s)$, for the system represented in Fig. 4.



3. Using Mason's rule, find the transfer function, $T(s) = C(s) / R(s)$, for the system represented by Fig. 5.

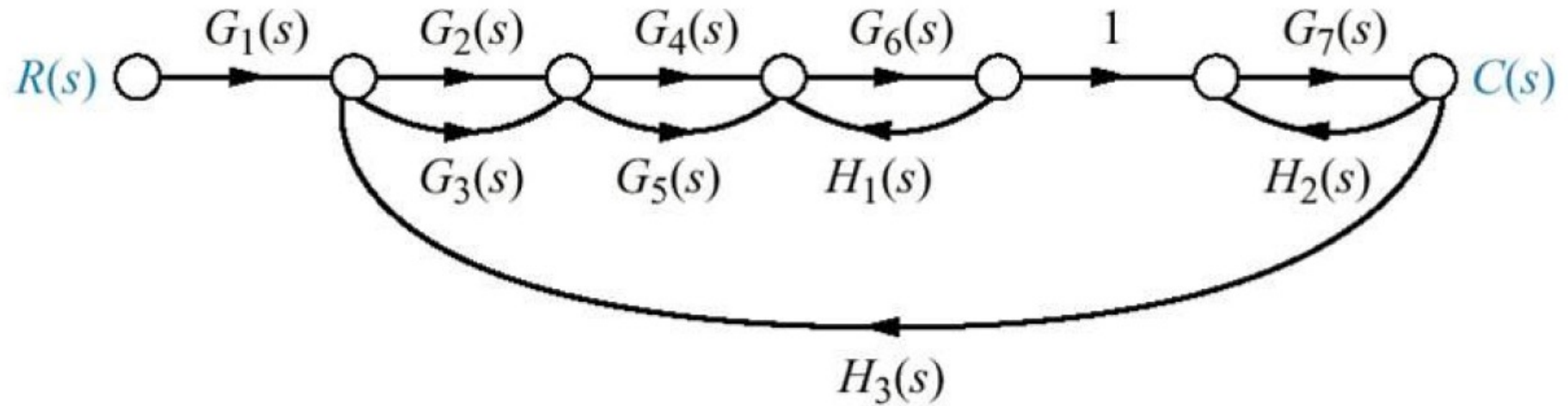


Fig. 5

Tuliskan :

1. Loop gain
2. Forward gain
3. Non-touching loop gain

$$G(s) = \sum \frac{T_k \Delta_k}{\Delta}$$

Jawab:

Loop gain:

$$L_1 = G_6 H_1$$

$$L_2 = G_7 H_2$$

$$L_3 = G_2 G_4 G_6 G_7 H_3$$

$$L_4 = G_2 G_5 G_6 G_7 H_3$$

$$L_5 = G_3 G_4 G_6 G_7 H_3$$

$$L_6 = G_3 G_5 G_6 G_7 H_3$$

Forward path:

$$F_1 = G_1 G_2 G_4 G_6 G_7$$

$$F_2 = G_1 G_2 G_5 G_6 G_7$$

$$F_3 = G_1 G_3 G_4 G_6 G_7$$

$$F_4 = G_1 G_3 G_5 G_6 G_7$$

Non-touching loop gain:

$$L_1 \text{ dan } L_2: G_6 H_1 G_7 H_2$$

$$G(s) = \sum \frac{T_k \Delta_k}{\Delta}, \Delta = 1 - (L_1 + L_2 + L_3 + L_4 + L_5 + L_6) + L_1 L_2,$$